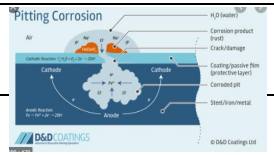
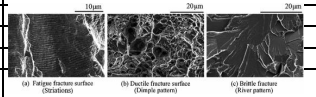
No	Date	FA Method	Item	Keywords (1)	Keywords (2)	Details	Review
	17/02/21	FA	FA	FlowChart		Sample In Visual Inspection/Non destructive test (Optical Microscope, SEM, X-RAY, CSAM) Sample Preparation (Mechanical cross-section, Ion milling cross cutting) Destructive test (cross-sectioning)	
	17/02/21	FA	FA	FA		Surface Analysis, Contamination, Composition, Imaging, Elemental Tech: Auger, SEM, EDS, FIB, FTIR, TOF-SIMS, XPS	
	17/02/21	Metallurgical				1. Fracture failure modes and origins. 2. Fatigue and other slow crack growth mechanisms 3.Brittle and ductile overload failures 4. Stress sorrosion cracking and hydrogen embrittlement failures 5.Corrosion and oxidation problems	
	17/02/21	Chemical				Elemental analysis Tech: Auger, XPS, EDS, ICP-OES, UV-VIS, AAS, FTIR, TOF-SIMS, XRF, U-XRF	
	18/02/21	X-Ray				2D/3D X-RAY	
	18/02/21	AAS				Atomic Absorption Spectroscopy (AAS)	
	18/02/21	AFM				Atomic Force Microscopy (AFM)	
	18/02/21	AES				AUGER Electron Spectroscopy (AES)	
	18/02/21	CSAM				Confocal Scanning Acoustic Microscopy (CSAM)	
	18/02/21	Decapsulation				Decapsulation / Decapping / Delidding >>Manual Chemical Etching: Remove the plastic material covering die. Cavity milled on the top surface of the package. Chemical Etching Recepi >>Mechanical Decap: >>Laser Decap:	
			Limitation			Sample Limitation	
						Alphabet not use for coordinate	
1	15/03/21	Xsect	Failure	Xsect	Scratches		
2	15/03/21	Xsect	Failure	Xsect	Pores		
3	15/03/21	Xsect	Failure	Xsect	Crack		
4	15/03/21	Xsect	Failure	Xsect	Gap		
5	15/03/21	Plating	Intro	Plating		>>Surface covering in which a metal is deposited on a conductive surface. to decorate objects, for corrosion inhibition, to improve solderability, to harden, to improve wearability, to reduce friction, to improve paint adhesion, to alter conductivity, to improve IR reflectivity, for radiation shielding, and for	
6	15/03/21	Plating	Type	Plating		>>Electroplating, Electroless plating. >>Gold plating, Silver Plating, Copper Plating, Rhodium Plating, Zinc Plating, Zinc-nickel plating, Tin plating, Alloy plating, Composite plating, Cadmium plating, Nickel plating, Electroless nickel plating (ENEPIG & ENIG)	
7	15/03/21	Plating	Compare	Coating		Coating and plating are techniques of surface covering. The key difference between coating and plating is that coating can be done on both conductive and non-conductive surfaces, whereas plating can be done on	
8	15/03/21	Plating	Process	Electroplating	PROCESS	Depend on the plating. ENIG (Electroless nickel immersion gold) Cleaning>>DI water rinsing>>Acid Cleaning>>DI water rinsing>>Soft Etching>>DI water rinsing>>Pre-dip>>PdCl2 activation>>DI water rinsing>>E'les Ni Plating>>DI water rinsing>>Immersion Gold	

9	15/03/21	Plating	Failure	Electroplating	Cold Shuts	Observation: flow marks or lines on the metal's surface. Cause: Metal has hardened at different stages Cold shuts are one of the most common defects found in materials sent for electroplating.	
10	15/03/21	Plating	Failure	Electroplating	PITTING	Observation: small holes on the surface of the object. Cause: chemical fume suppressants, Very small holes, oxidation on the surface, microscopic fissures and pieces of non-metallic materials can also cause pitting. Finally, pitting may be caused by stop-off residue, the breakdown of tank plumbing or lining, wire brush debris, grit from a blaster, buss arching and substances like oil	
11	15/03/21	Plating	Failure	Electroplating	SHARP EDGES	Observation: Sharp Edges Cause: process itself relies on current, which starts a reaction on the part's surface and attracts cations. Certain geometrical shapes will result in more attraction than others.	
12	15/03/21	Plating	Failure	Electroplating	CLEAVAGE POINTS	Observation: splitting of a solid along a structural plane. Cause: die casting process.	
13	15/03/21	Plating	Failure	Electroplating	Cracking	CRACKING AFTER PLATING — HYDROGEN CRACKING Solution: Enacting [put into practice] several measures throughout the process. These include stress relief baking and peening the object with shot, both of which promote strength and minimize the chance of hydrogen	
14	15/03/21	Plating	Failure	Electroplating	DULL AND HAZY	DULL AND HAZY DEPOSITS IN PLATING **Chemical Imbalance:** When immersing the object in a bath, insufficient levels of Dura Additive or an excessive level of sulfate, chromic acid or contaminants can lead to dull or hazy deposits in the final product. **Incorrect Temperature:** The bath temperature needs to be between 130 and 140 degrees Fahrenheit. Use an air agitator to keep the temperature consistent throughout the bath, and try to stay within a margin of two degrees Fahrenheit. **Incorrect Current Density:** Current density should be two ASI, and deviating from this level can cause dullness or haziness. Incorrect Distribution of Current: If the anodes are not the proper distance from the object, the current will not flow correctly. Use conforming anodes to help with this problem. **Improper Rectifier Function**: The rectifier must be working properly to avoid dull deposits. Perform tests to look for needed repairs. **Inadequate [insufficient] Preheating:** Parts that are not hot enough can wind up with defects. Ensure that they have heated up to the bath temperature all the way through. **Interrupted Current:** Parts that are plated without current interruption tend to have the best results.	
15	15/03/21	Plating	Failure	Electroplating	BLISTERING	BLISTERING IN PLATING Cause: These gases are usually either hydrogen or nitrogen	
16	15/03/21	Plating	Failure	Electroplating	OXIDATION	Effect: lead to poor adhesion and a lifting of the plating after the process is complete.	
17	15/03/21	Plating	Failure	Electroplating	ENIG	Characterization of the Contamination Factor of Electroless Ni Plating Solutions on the ENIG Process https://link.springer.com/content/pdf/10.1007/s11664-018-6335-1.pdf -often produces defects, such as pinholes and black pads that can cause brittle fractures at the interface	
18	15/03/21	Plating	Failure	Electroplating	pinholes		
19	15/03/21	Plating	Failure	Electroplating	black pads		
20	15/03/21	Plating	Failure	Electroplating	brittle fractures		
21	15/03/21	Plating	Failure	Electroplating	white patches	nickel plating very high chance it's due to organic contamination in nickel bath. You can run a Hull Cell test to see if similar white patches appear on hull cell panel after plating. If so, do a carbon treatment (add in 2 g/l activated carbon powder and maintain at ~50 °C with circulation for 1 hour) of same sample and re-run hull cell again and if such	
		Apple				Electrical Fault Isolation	
		Apple				FA using X-ray/SEM/CSAM/FIM/X-sect	
		Apple				Surface morphology tools	
		Apple				IC assembly packaging design, process and testing	
		Apple				Auger Experience is a plus	
		Apple				Nano-indenter experience is a plus	
		Apple				Process/FA experience in semiconductor or optical devices is a plus	
		Apple					
		Apple				dynamic and ever-changing work environments	
		Apple				Strong solution-focused work ethic with a high degree of flexibility supporting multiple projects of varying and priority	
		Apple				Self motivated with strong written and verbal communications skills	
		Apple				Provide comprehensive FA during NPI process Work with various functional teams for root cause identification, simulation and solution	

				PCB	Solder voids	Solder with voids (Koko) Coating uneven https://www.intechopen.com/books/welding-modern-topics/overview-of-selected-issues-related-to-soldering	
				PCB	Solder voids	Causes of microvoids https://www.assemblymag.com/articles/84024-avoiding-microvoids insufficient temperature during reflow. outgassing of organic brighteners within the underlying copper deposit. Impure flux in the solder paste. Volatilization of water on the PCB or within the solder paste. Excess oxides on the silver deposit. Solder mask contamination on the copper.	
				PCB	Solder voids	Gold Coating spike	
				PCB/Coating	Al Corrosion	Aluminium corrosion Class Notes Electrochemical Series	
				Metalalogy		metal. https://www.researchgate.net/figure/Constituent-zones-of-a-fusion-welding-1-Weld-metal-zone-2-Fusion-boundary-3-Heat_fig5_331106038 Characterization of fusion welded joint: a review	
				XRAY-CSAM		XRay Scan metal, SCAM scan all material with different hardness, must know its design.	
				Epoxy x-sect		for small gap. using slow cure exposy and put in vacuum, will 2nd mould if needed.	
				XRAY		XRAY Important Point XRAY Cannot detect Aluminium Wire	
				Solder		Type of Solder and its composition	
				FA Test		FA test and its function	
				Coating failure		Rose Case/wiedland case	
				FET Crystal TEM		Analysis FET crystal	
				Plating		Ni Plating	
	23/03/21	SEM				drifting because of material design	
	23/03/21	IM				Solder ball melt, becuase temp too high	
	24/03/21	EDX				SDD	
	24/03/21	EDX				Smearing while mechanical x-section	
				Die Casting			
				Metal Casting			

	15/03/21	Coating	Compare	Plating		Coating and plating are techniques of surface covering. The key difference between coating and plating is that coating can be done on both conductive and non-conductive surfaces, whereas plating can be done on	
		FTIR	Technique	FTIR	Mixture	Quantitative analysis	
5							
6							
7							
8		EBSD		EBSD	Grain Boundary		
9					Phase		
10					Strain		
11					Misorientation		
12					IPF	Inverse Pole Figure	
13							
14				FPGA			
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16							
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18							
19						Isolation	
20		Keyword	Keyword	Keyword	remedy	Medicine/Solution	
21		Keyword	Keyword	Keyword	reliability	the quality of being trustworthy or of performing consistently well.	
22		END					

Useful Link

https://incompliancemag.com/article/failure-analysis-a-road-map-2/

No	Date	FA Method	Type
1	17/02/2021	FA FlowChart	
2	17/02/2021	FA	
3	17/02/2021	Metallurgical	
4	17/02/2021	Chemical Analysis	
5			
6	18/02/2021	X-Ray	
7		AAS	
8		AFM	
9		AES	
10		CSAM	
11		Decapsulation	
12			
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23	27/01/2021	EDX	Theory
24	27/01/2021	EDX	Theory
25			Limitation
26			
27			
28			FA Analysis
29			FA Analysis
30			
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36			
37	27/01/2021	PCB	Solder voids
38	27/01/2021	PCB	Solder voids
39	27/01/2021	PCB	Solder voids
40	17/02/2021	PCB/Coating	Al Corrosion
41	17/02/2021	Metalalogy	
42	19/02/2021	XRAY-CSAM	
43	19/02/2021	Epoxy x-sect	
44	03/04/2021	XRAY	
45	03/04/2021	Solder	
46	03/04/2021	FA Test	
47	03/04/2021	Coating failure	
48	03/04/2021	FET Crystal TEM	
49		Plating	
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QnA
Sample In Visual Inspection/Non destructive test (Optical Microscope, SEM, X-RAY, CSAM) Sample Preparation (Mechanical cross-section, Ion milling cross cutting) Destructive test (cross-sectioning)
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2D/3D X-RAY
Atomic Absorption Spectroscopy (AAS)
Atomic Force Microscopy (AFM)
AUGER Electron Spectroscopy (AES)
Confocal Scanning Acoustic Microscopy (CSAM)
Decapsulation / Decapping / Delidding >>Manual Chemical Etching: Remove the plastic material covering die. Cavity milled on the top surface of the package. Chemical Etching Recepi >>Mechanical Decap: >>Laser Decap:
EDX Sensitivity (two spectrum energy near)
Type of Auto ID
Isolation
Qualification
Quantification
Solder with voids (Koko) Coating uneven https://www.intechopen.com/books/welding-modern-topics/overview-of-selected-issues-related-to-soldering
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Gold Coating spike
Aluminium corrosion Class Notes Electrochemical Series

Constituent zones of a fusion welding: 1. Weld metal zone, 2. Fusion boundary, 3. Heat affected zone, 4. Parent metal.
https://www.researchgate.net/figure/Constituent-zones-of-a-fusion-welding-1-Weld-metal-zone-2-Fusion-boundary-3-Heat_fig5_331106038

Characterization of fusion welded joint: a review
https://www.semanticscholar.org/paper/Characterization-of-fusion-welded-joint%3A-a-review-Lah-Ali/bff4af2d586444a7d4a069805f1d0b536013e8da

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